Fiscal Year:	FY 2012	**Task Last Updated:**	FY 02/01/2012
PI Name:	Durian, Douglas J. Ph.D.		
Project Title:	The Melting of Aqueous Foams		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	19104-6396	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	1995 Microgravity Fluid Physics NRA-94-OLMSA-05: Research and Flight Experiment Opportunities
Start Date:	08/05/2004	**End Date:**	04/04/2013
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Motil, Brian J.	**Contact Phone:**	216-433-6617
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Flight Program:			
Flight Assignment:	NOTE: Grant number was changed in 2007 to NNX07AP20G (Ed., 1/4/13) NOTE: Funding extended to 4/04/2013 per NSSC database (5/2010) NOTE: Funding extended to 4/4/2010 through a new grant in 2007 (6/2008) NOTE: Funding reinstated thru 9/30/2008, per F. Chiaramonte (8/2006). Suppl funding thru Oct 2006 (6-5-06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX07AP20G ; NNX07AK60G ; NNC04GB61G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The unusual elastic character of foams will be quantified macroscopically by measurement of the shear stress as a function of static shear strain, shear strain rate, and time following a step strain; such data will be analyzed in terms of a yield stress, a static shear modulus, and dynamical time scales. Microscopic information about bubble packing and rearrangement dynamics, from which these macroscopic non-Newtonian properties presumably arise, will be obtained non-invasively by novel multiple-light scattering diagnostics such as diffusing-wave spectroscopy (DWS) and speckle-visibility spectroscopy (SVS). Bubble size and homogeneity will be studied by diffusely transmitted light (DTS). Liquid fraction will be measured by electrical conductivity. Quantitative trends with materials parameters, such as average bubble size, and liquid content, will be sought in order to elucidate the fundamental connection between the microscopic structure and dynamics and the macroscopic rheology. There are three all-encompassing objectives motivating the FOAM project. The first concerns multiphase flows and represents the “mechanics” aspect alluded to in the F.O.A.M. acronym: To understand how foams and bubbly liquids deform and flow in response to applied forces. The second objective concerns multiphase separations: To understand how foams and bubbly liquids evolve with time via a combination of liquid flow between bubbles, of gas diffusion across the film separating neighboring bubbles, and of direct coalescence via film rupture. For both flows and separations, some level of understanding has been established for very dry foams, and also for liquids with relatively few bubbles; however, we are more ignorant of the crucial intermediate regime of very wet foams and very bubbly liquids. This regime defies earth-bound experiments. The third, overall, objective is to transfer this new understanding of multiphase flows and separations into ASL technology for NASA's strategic exploration mission. The utility and fascination of foams are derived largely from the surprising fact that they have a solid-like elastic character in spite of being mostly gas with a few percent volume fraction of liquid, but can nevertheless flow under shear. The physical origin of such unusual rheology in terms of microscopic structure and dynamics is poorly understood and remains a subject of basic scientific interest to physicists, chemists, and chemical engineers. The proposed research promises important new insight into these issues, and could also have significant consequences for our understanding of flow in other dense randomly-packed systems such as emulsions, colloidal suspensions, slurries, bubbly liquids, and granular materials. Furthermore, all foam applications are empirically based and the proposed research may generate valuable fundamental guidance for the development of materials with more desirable rheology and better stability. The value of the FOAM project in terms of NASA’s exploration mission lies with Advanced Life Support. Water recovery and recycling are absolutely central to ASL, and require fine control of multiphase flows and separations. The very heart of the FOAM project is to bring new technology and understanding to these problems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Foams are important for a clean environment. Firstly, their presence in natural waters signals the presence of surface-active impurities and a need for clean-up. Secondly, they are useful themselves in the cleanup process since many impurities can be removed by injecting gas and skimming off the resulting foam. Relatedly, study of foams brings specific insight into multiphase flow and separation phenomena of direct importance for water recovery and recycling operations.
Task Progress:	During this period my group consisted of only Penn graduate student Adam Roth. During the past year we have made significant progress on several different aspects of the overall FOAM project, all pertaining to coarsening and with a view toward supporting the planned flight experiments: The first major effort was to analyze image data for the coarsening rates and the bubble-scale topology and shape characteristics -- all as a function of liquid fraction. This involved coming to the realization that it's the area-weighted distributions that matter most, rather than the number-weighted distributions that have been the focus of prior work for 2d dry foams. After elucidating this physics, we analyzed data for the relevant new quantities and discovered that many features are independent of liquid fraction. In addition to acquiring record-best statistics on other, more usual, quantities, we also documented the deviation of the coarsening rate from von Neumann's law for progressively wetter foams. And we successfully modeled this behavior. The results will serve to anticipate features expected in flight experiments on progressively wetter 3d foams. And finally we have written up the results and polished the manuscript to the point that submission will happen very soon. In addition to varying the wetness of 2d foams, we also complete an experiment where the geometry of the embedding space is curved. This alters the von Neumann law in a way predicted some time ago but never tested in experiment. Tasks included completing construction of a sample cell, where the foam is confined between concentric clear Lucite domes; developing an image processing procedure to eliminate the "double image" of the borders of the bubbles which are away from the north pole, at the top of the dome, where the camera is located; taking and analyzing data; and completing a rather-rough draft of a manuscript. One result worth noting is that the area of a six-sided bubbles grows with time, at a rate that depends on on area, in accord with the Avron-Levine prediction. By contrast, in 2d flat space, six-sided bubbles do not change area at all. In a third area, we began a new series of experiments on the evolution and shape statistics of edge bubbles of dry 3d foams. We wish to discover if the distributions are stationary, and if they depend upon liquid content. On earth the liquid content can be varied only minimally when homogeneous. Large gradients and correlations with bubble size necessarily occur if the foam is very wet; hence the need for extended microgravity conditions in the upcoming flight experiments. In yet another on-going project, we returned to analyzing tomography data collected for very dry 3d foams.
Bibliography Type:	Description: (Last Updated: 06/12/2025)
Abstracts for Journals and Proceedings	Chieco A, Feitosa K, Roth AE, Korda PT, Durian DJ. "Digital reconstruction of 3d polydisperse dry foam." Annual March Meeting of the American Physical Society, Boston, MA, February 27-March 2, 2012. Bulletin of the American Physical Society 2012 Mar;57(1):C1.00118. http://meetings.aps.org/Meeting/MAR12/Event/160135 , Mar-2012

Abstracts for Journals and Proceedings	Roth AE, Jones CD, Durian DJ. "Coarsening of two dimensional foams on a curved surface." Annual March Meeting of the American Physical Society, Boston, MA, February 27-March 2, 2012. Bulletin of the American Physical Society 2012 Mar;57(1):A51.00008. http://meetings.aps.org/Meeting/MAR12/Event/159280 , Mar-2012
Articles in Peer-reviewed Journals	Sebag V, Roth AE, Durian DJ. "Final bubble lengths for aqueous foam coarsened in a horizontal cylinder." Philosophical Magazine 2011 Dec1;91(34):4357-66. http://dx.doi.org/10.1080/14786435.2011.606239 , Dec-2011
Books/Book Chapters	Dauchot O, Durian DJ, van Hecke M. "Dynamical heterogeneities in grains and foams." in "Dynamical heterogeneities in glasses, colloids, and granular media." Ed. L. Berthier et al. Oxford ; New York : Oxford University Press, 2011. http://w3.lcvn.univ-montp2.fr/~lucacip/DH_book.htm#_Chapter_6:_Dynamical , Jul-2011